Connecting the Drone to Mission Planner

Drone Info

Connection Info:
static ip_address=192.168.1.100/24 (you can find with photonflight.local; subnet mask is 255.255.255.0)
Port: 5678

Use RX4&TX4 on the flight controller side.

Pi needs to be connected to Vx and Gnd for power; reason being that Vx can supply more current than the other pins.
Currently, the Pi is connected to serial2 on the flight controller

Credentials
Username: photonflight
Password: *rZ866Qr8!IRnr

Power:
Vx and Gnd from Servo side of the flight controller board

Camera Code Required:

```
rpicam-vid -t 0 \ --width 1920 --height 1080 --framerate 30 \ --codec h264
--profile high --level 4.1 --inline \ --post-process-file
/usr/share/rpi-camera-assets/imx500_mobilenet_ssd.json \ -o - \ |
gst-launch-1.0 fdsrc \ ! h264parse \ ! rtph264pay config-interval=1 pt=96 \ !
udpsink host=192.168.1.155 port=1110
```

Prof. Burke's install script: https://github.com/mavlink-router/mavlink-router

Mission Planner Info

Network Settings:
Laptop IP: 192.168.1.155
Subnet Mask: 255.255.255.0
Connect via TCP to port 5678

Firewall Settings:
Go to "Network & Internet" -> "Advanced Network Settings" -> "Advanced Sharing Settings"
Turn on Network Discovery inside of Private Networks

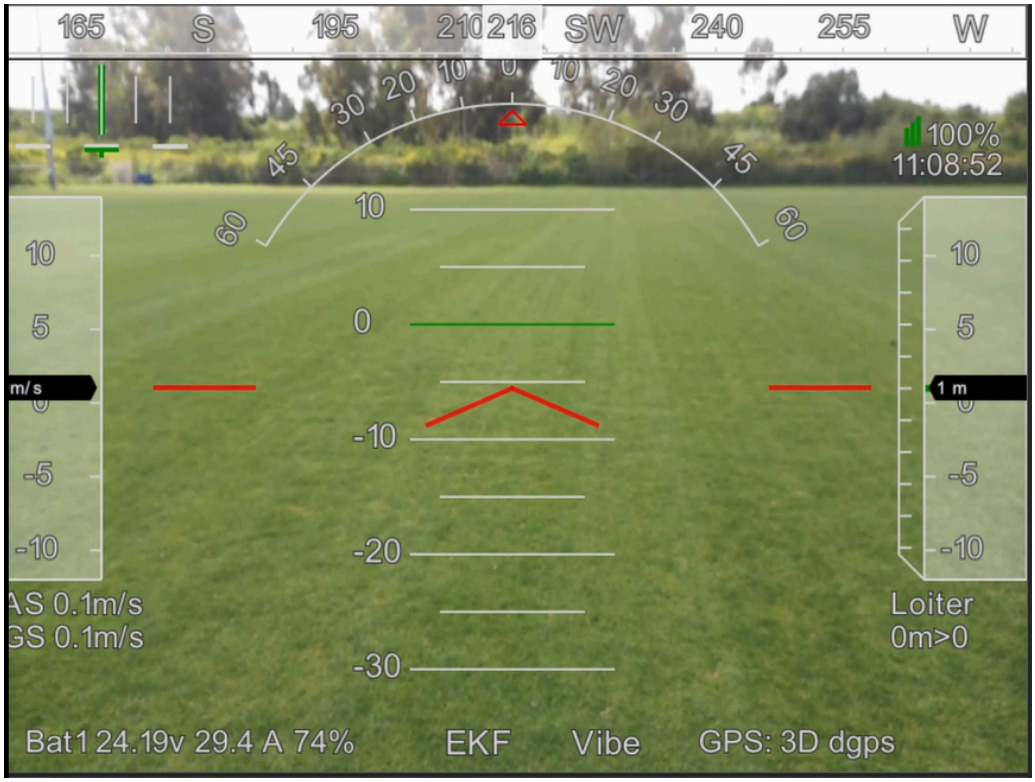
Camera Code Required:

```
udpsrc port=1110 caps="application/x-rtp, media=video,
```

```
encoding-name=H264, payload=96" ! rtph264depay ! h264parse ! avdec_h264 ! queue max-size-buffers=1 max-size-bytes=0 max-size-time=0 leaky=2 ! videoconvert ! video/x-raw,format=BGRA ! appsink name=outsink sync=false
```

Instructions

1. Connect drone to laptop via fiber optics and ethernet
    a. Raspberry pi -> ethernet -> SFP on drone
    b. SFP on drone -> fiber -> SFP on laptop
    c. SFP on laptop -> ethernet -> laptop
2. **For camera make sure it is connected before the Pi gets power. Otherwise a reboot is required.**
3. First turn on drone by connecting main battery
4. Wait for flight controller to beep and for raspberry pi to turn on and run scripts
    a. To check to see if the Pi is running by pinging 192.168.1.100 in the terminal
5. On Mission Planner select TCP and connect to 192.168.1.100 at port 5678
6. Mission Planner should be pulling data over the wired connection and relaying that information back to the laptop
7. **Joystick Config:**
    a. Then connect the radio controller to Mission Planner via USB type C cable
    b. Make sure radio controller is running in wired joystick mode
    c. Go to setup -> joystick and then configure the controller
8. **If using Fiber:**
    a. To start video:
        i. Right Click on HUD
        ii. Hover over “Video”
        iii. Click “Set Gstreamer Source”
        iv. Paste in string from Mission Planner Info
        v. Hit Enter, should get a feed onto the HUD

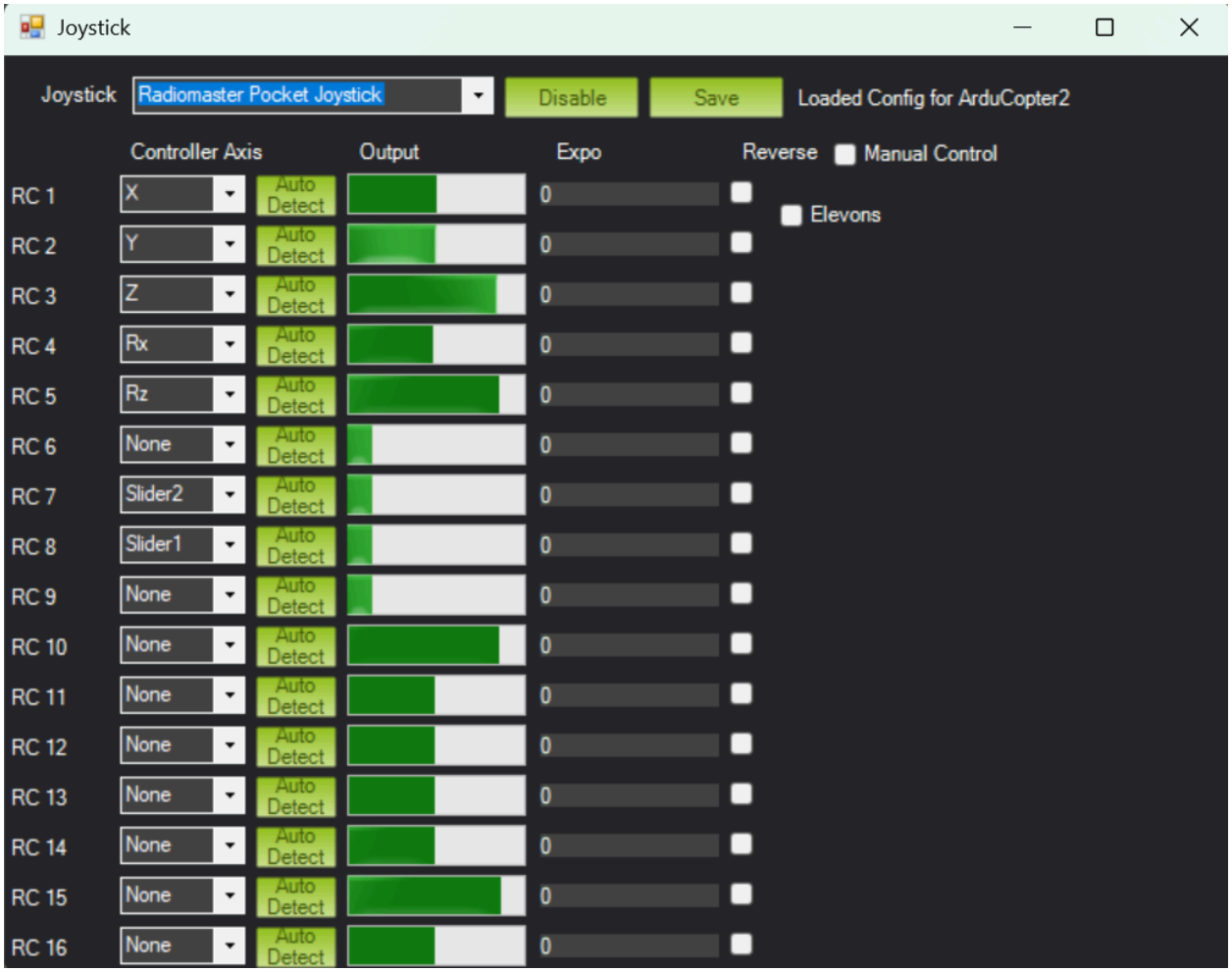


Findings (1-28-26)

- We hooked up RX & TX on the flight controller side to Serial 4. Checked the parameter file and saw that baud rate & communication protocol weren't set correctly
- Changed the baud rate to 57600 (57 in settings), and changed protocol to MavLink2 (2 in settings)
- On Pi Tx4 is plugged into GPIO 15 (Rx)
    - Rx4 is plugged into GPIO 14 (Tx)
- After rebooting flight controller we are able to connect without issues

Joystick Config

1. Setup -> Optional Hardware -> Joystick
2. https://ardupilot.org/copter/docs/common-joystick.html
3. The channels that Burke setup are going to match the RC #s listed in the setup menu
    a. EX: RC1 deals with Aileron which means rolling left and right (tilt). So we hit auto detect and move the stick that deals with rolling left and right. This happens to be the one that maps to the X axis of the controller



As of 1-28-26

Turning On The Camera

This is assuming the connection has been setup and that you have SSH'd into the Pi.

1. First navigate to the folder with the code:

cd Projects/EECS-159A-Drone/Vision\ Tracking\ pipeline/

2. Activate the venv (Virtual Environment):

source .venv/bin/activate

3. You should now see a (.venv) in green text on the left of your terminal
4. Execute the program:

python3 Object_Tracking.py

5. Now the program will first:
    a. Connect to the drone (using the same port as mission planner 5678)
    b. And then upload firmware to the camera (this is when the loading bar appears)
    c. A menu will print out as the loading bar appears and then will get pushed up by more text
    d. TLDR it should look like this:

```
------------------OPTIONS MENU------------------
Type the option and hit enter.
1. Enable Detection and Drone Tracking
2. Enable AI Camera Feed
q. Quit the Program
------------------------------------------------
```

6. Select an option (use #2 if you do not want the drone to react to objects)
    a. Press 1 or 2 and hit enter
    b. Drone must be armed before it can react
    c. 1 and 2 will start object detection on Mission Planner (assuming you've already set that up by pasting the long string into Gstreamer)
7. When you want to quit press: Ctrl + C
8. That will exit you back out to the menu; the camera glitching is normal because the program will restart the camera
9. To fully quit the menu press Ctrl + C again or press q and hit Enter
10. **If you want to rerun the program (assuming you haven't done anything else) just go back to step #4**